



Time = 25 Minutes

Quiz No 10

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly.

Most of the total score for the questions will be awarded on your reasoning.

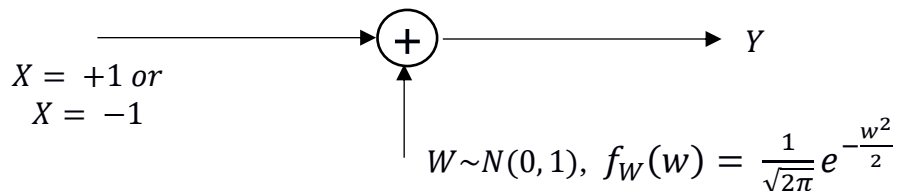
CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [20 points] [Bayes Rule - Inference]

Random Variable X is discrete and **Y is continuous**. The question pertains to the case (as discussed in the class) where we can observe **Y** and would like to infer **X** using the conditional probability, $P_{X|Y}(x|y)$. Assume that we know the prior distribution of **X**, $P_X(x)$ and the model $f_{Y|X}(y|x)$.

Part a [06 points] Derive the expression $P_{X|Y}(x|y)$. All necessary steps are to be shown.

Now assume the following system (see below) pertaining to the above-mentioned case that involves the discrete **Random Variable X** and a continuous Random Variable, **Y**. This is a simple model for a communication system.



Part b [07 points] Given $P(X = +1) = P(X = -1) = 0.5$, find and plot $P_{X|Y}(+1|y)$. Interpret the plot with regards to estimating **X**.

Part c [07 points] Now assume that $P(X = +1) = 0.75$ and $P(X = -1) = 0.25$. Find the revised $P_{X|Y}(+1|y)$ and plot it. Interpret the plot with regards to estimating **X**.

Note – provide necessary justifications.